人工智慧無線通訊
Artificial Intelligence Wireless
HW-2

Exercise 2.1

Consider the set of empirical measurements of *Pr/Pt*, i.e., the ratio of received to transmitted power given in Table 2.4 at 900 MHz.

Find the path loss exponent y and the shadowing variance in the simplified path loss and shadowing model in (2.1) that best fit the data in terms of minimizing the MSE, assuming d_0 = 1 m and K is determined from the free-space path loss model at this do given by $10\log_{10} K = 20\log_{10}\frac{\lambda}{4\pi d_0}$.

$$PL(d) = 10\log_{10} K + 10\gamma\log_{10}\left(\frac{d}{d_0}\right) + \phi_{dB}$$

Step1. Compute *K* from the Free-Space Model:

Given frequency

$$f = 900\,MHz \rightarrow$$

wavelength

$$\lambda = \frac{3\times 10^8}{9\times 10^8} = 0.333m$$

$$10\log_{10} K = 20\log_{10}\left(\frac{\lambda}{4\pi d_0}\right) \rightarrow 10\log_{10} K = 20\log_{10}\left(\frac{0.333}{4\pi}\right)$$

$$\rightarrow 10\log_{10} K = -31.54\,dB$$

Step2. Path Loss Model:

Since $d_0 = 1$

$$PL(d) = -31.54 + 10\gamma\log_{10}(d) + \phi \rightarrow$$

Ignoring the shadowing term for fitting

$$PL(d) \approx -31.54 + 10\gamma\log_{10}(d)$$

Step3. Linear Regression Form

Let

$$x = \log_{10}(d)$$
$$y = PL(d)$$

$\rightarrow$

Then the model becomes

$$y = a + bx$$

$\rightarrow$

$$a = -31.54$$
$$b = 10\gamma$$

Step4. Regression Table

d (m)	*PL(dB)*	*$\log_{10}(d)$*
5	-60	**$\log_{10}(5)$** = 0.699
30	-80	**$\log_{10}(30)$** = 1.477
60	-105	**$\log_{10}(60)$** = 1.778
110	-115	**$\log_{10}(110)$** = 2.041
500	-135	**$\log_{10}(500)$** = 2.699

Step5. **Least Squares for Exponent**

$$\gamma = -\frac{\sum (M_i + 31.54) x_i}{\sum x_i^2}$$

x	y	$x - \bar{x}$	$y - \bar{y}$	x * y	$(x - \bar{x})2$
0.699	-60	-1.040	39	-40.56	1.082
1.477	-80	-0.262	19	-4.98	0.069
1.778	-105	0.039	-6	-0.23	0.002
2.041	-115	0.302	-16	-4.83	0.091
2.699	-135	0.960	-36	-34.56	0.922

$$\gamma = -\frac{-6715.4}{1727.8}$$

$$\gamma = 3.88$$

Step6. **Path Loss Model**

$$P_r/P_t\ (\text{dB}) = -31.54 - 38.8 \log_{10}(d)$$

Step7. **Shadowing Term**

Distance (m)	Measured PL (dB)	Model PL (dB)	Error
5	-60	-58.65	1.82
30	-80	-88.84	78.15
60	-105	-100.52	20.07
110	-115	-110.72	18.32
500	-135	-136.25	1.56

Final Answer:

Path loss exponent

$$\gamma \approx \mathbf{3.88}$$

Shadowing Variance:

$$\sigma^2_{\psi_{\text{dB}}} = \frac{119.92}{5} \approx \mathbf{23.98}\ \text{dB}^2$$

Exercise 2.2

Step 1. Mean Delay

$$\bar{\tau} = \sum P_i \tau_i$$

$$\bar{\tau} = 0.61722(0) + 0.30934(50) + 0.06172(110) + 0.00978(170) + 0.00155(290) + 0.00039(310)$$

$$\bar{\tau} \approx 24.49 \text{ ns}$$

Step 2. Second Moment

$$E[\tau^2] = \sum P_i \tau_i^2$$

$$E[\tau^2] \approx 1970.66$$

Step 3. RMS Mean Delay

$$\tau_{rms} = \sqrt{E[\tau^2] - \bar{\tau}^2}$$

$$\tau_{rms} = \sqrt{1970.66 - (24.49)^2}$$

$$\tau_{rms} \approx 37 \text{ ns}$$

Step 4. Symbol Duration Condition

$$T_s \geq 10\tau_{rms}$$

$$T_s \geq 10(37)$$

$$T_s \geq 370 \text{ ns}$$

Step 5. Maximum Symbol Rate

$$R_s = \frac{1}{T_s}$$

$$R_s = \frac{1}{370 \times 10^{-9}}$$

$$R_s \approx 2.7 \text{ Msymbol/s}$$

Final Answer:

RMS Delay Spread	Maximum Symbol Rate
$\tau_{rms} \approx 37 \text{ ns}$	$R_s \approx 2.7 \text{ Msymbol/s}$

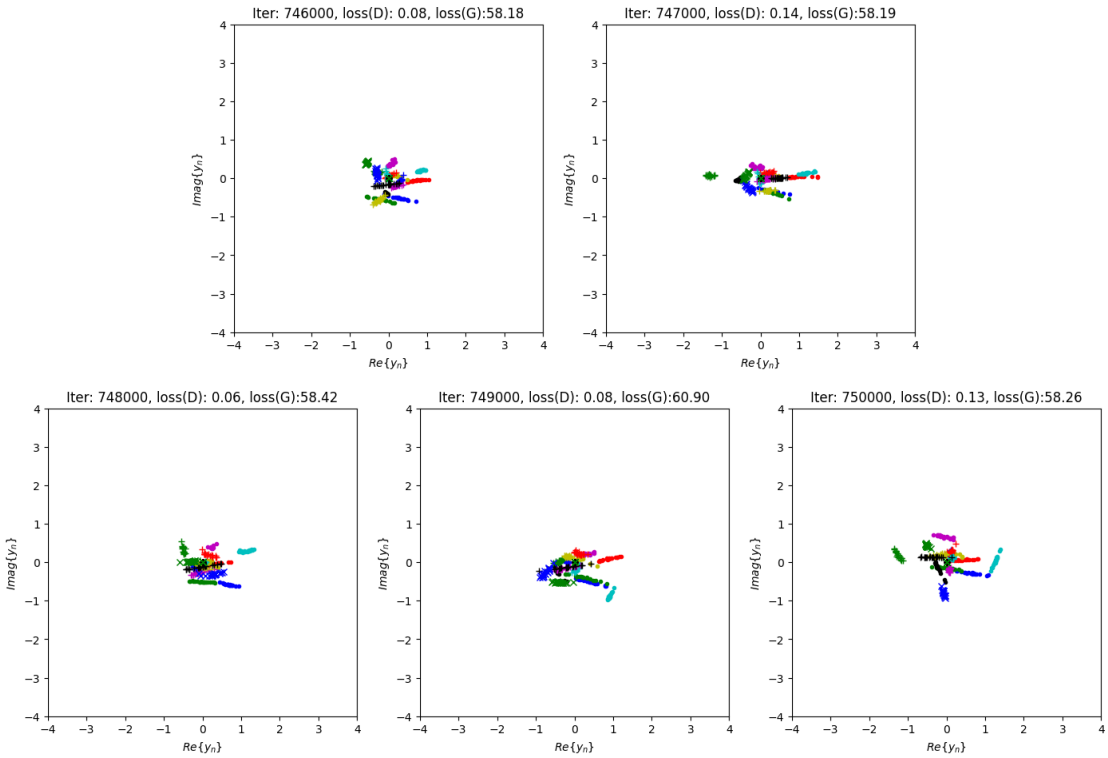
Exercise 2.4

QUAsi Deterministic RadIo channel GenerAtor (QuaDRiGa), is a powerful toolbox for generating realistic radio channel impulse responses for system-level simulations of mobile radio networks. QuaDRiGa was mainly developed to enable the modeling of MIMO channels for specific network configurations, such as indoor, satellite, or heterogeneous configurations. In addition, QuaDRiGa contains a collection of features createdin SCM(e) and WINNER channel models along with novel modeling approaches that provide features to enable quasi-deterministic multi-link tracking of user (receiver) movements in changing environments.

(a) Use QuaDRiGa to generate real MIMO channels for one specific configuration.
https://github.com/ZeinyLing/Artificial-Intelligence-Wireless/blob/main/HW-2/rayleigh_channel_dataset.mat

(b) Use the GAN-based framework to model the MIMO channel you generated in 2.4(a).

The results generated in the last 5000 iterations



Code: https:/github.com/ZeinyLing/Artificial-Intelligence-Wireless/blob/main/HW-2/Exercise_2.4_starter.py